



ries, and beyond. The sensor ensures user privacy by not capturing images, distinguishing it from commonly used camera sensors for similar purposes.

STMicroelectronics  
<https://www.st.com>

## Smallest Smart Waste Sensor

Adhoc networks have launched the world's smallest smart waste sensor module, OSCAR, measuring 71x43x28mm. It measures container fill level up to 4m with a 27° cone angle. These modules enable automated



fill-level detection, reducing unnecessary collection trips for local authorities and companies. The optical sensor monitors waste container fill level, enabling efficient collection scheduling and reducing collection fleet CO2 emissions by up to 40%. With its low power consumption, the battery can operate for up to five years, based on twelve daily fill measurements and three data uplinks to the cloud. Due to its compact size, the sensor can be used in small containers, such as public trash cans.

Adhoc networks  
<https://ad-hoc.com>

## Powerful Wi-Fi HaLow Module

Silex Technology has launched SX-SDMAH, an industrial-grade Wi-Fi highly efficient, long range (HaLow) secure digital input/output (SDIO) module that offers high transmission



power and maximises long-range, high data throughput capabilities. The module adds flexibility to embedded system

design, ensuring Wi-Fi HaLow device interoperability and enabling approved countries to utilise up to 8MHz bandwidth. Additionally, it can withstand an industrial temperature range of -40°C to 85°C. The surface-mount small form-factor module measures 17mm x 18mm x 2.65mm, complete with an integrated

## Stop Press!

ButtonBoard LLC, with EFY Labs' R&D support, launches an AI-enabled development board, revolutionising digital interactions. It is India's first small circuit board measuring 3cm in length and surpasses other boards in its segment in terms of capabilities. The board includes features like over the air (OTA) wireless connections for IoT and wearable design, offering a wide range of impressive projects. The board facilitates the integration of ChatGPT via application programming interface (API), along with bundled software that empowers developers and creators to embark on AI projects and implement assistance on this compact board. The ButtonBoard stands apart from other boards and kits by offering built-in sensors and a power voltage regulator, ensuring efficient power management.

ButtonBoard LLC  
<https://sites.google.com/view/buttonboards/home?authuser=0>

MHF-1 antenna connector, ensuring easy integration into various applications.

Silex Technology  
<https://www.silxtechnology.com>

## Innovative AI/ML Hardware in Sub-GHz SoC

Silicon Labs has launched an advanced FG28 system-on-chip (SoC), a dual-band solution designed for long-range networks, supporting protocols like



Amazon Sidewalk, wireless smart utility network (Wi-SUN), and other proprietary protocols. It is the industry's first integrated artificial

intelligence/machine learning (AI/ML) hardware accelerator within a sub-GHz SoC. The SoC enables on-device machine learning inference for applications such as predictive maintenance warnings and monitoring soil conditions. It offers ample storage with 1024 kB of flash storage and 256 kB of RAM to cater to the memory requirements of various protocols and technology stacks.

Silicon Labs  
<https://www.silabs.com>

## World's First Integrated Wi-Fi RF FEM for Wi-Fi 7 Access Points

Broadcom has launched film bulk acoustic resonator (FBAR) integrated front-end module (FiFEM) devices for Wi-Fi access points. The devices offer

5GHz and 6GHz band coexistence, low insertion loss, and significantly reduced bill of materials (BOM) at the



RF front end. They feature an advanced power amplifier design, optimising power consumption by up to 40%. The 6GHz Wi-Fi brings faster speeds, lower latency, and increased capacity worldwide, benefiting businesses and consumers.

Broadcom  
<https://www.broadcom.com>

## World's First Optical Network-on-Chip Processor

Hummingbird is the world's first optical network-on-chip (oNOC) processor launched by Lightelligence. This technology revolutionises big data interconnects for domain-specific artificial



intelligence (AI) workloads. The processor uses

stacked packaging to integrate photonic and electronic chips, serving as a communication network for data centres and high-performance applications. Hummingbird's waveguides transmit signals at the speed of light, enabling an all-to-all data broadcast network to each core on a 64-core AI processor. This gives a latency and power advantage over traditional digital interconnects.

Lightelligence  
<https://www.lightelligence.ai>